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ECEN 3730 – PCB Design Manufacture
09/30/25

Board 1 Report

Objective:

The objective of developing board 1 is to understand the 555-timer and its output. The circuit is powered by 5V to produce a timer output of 500Hz with a 66% duty cycle. It was designed in Altium using the provided library with multiple 1206 components. It includes three probe points. One to measure the 5V power, a second to measure the 555-timer's output, and a third to measure the load across an LED with a 1kΩ resistor. Using the timer's output, four LEDs are placed at the end. Each with a corresponding resistor of 10kΩ, 1kΩ, 300Ω, or 47Ω. Two switches are included throughout the design. One between the 5V power and the rest of the circuit as well as one between the timer and the LEDs.

Project Overview:

1. The circuit must be designed correctly in Altium passing all Design Rule Checks.
2. Must include an LED to show indicate power is on.
3. Must include a 555-timer producing a square output of approximately 500Hz with an approximate 66% duty cycle.
4. Must include four LEDs, each driven by a different resistor value.
5. Must include switches to isolate power and signals.
6. Must include a test point on one of the LEDs.
7. All components must be soldered neatly and correctly as good practice.

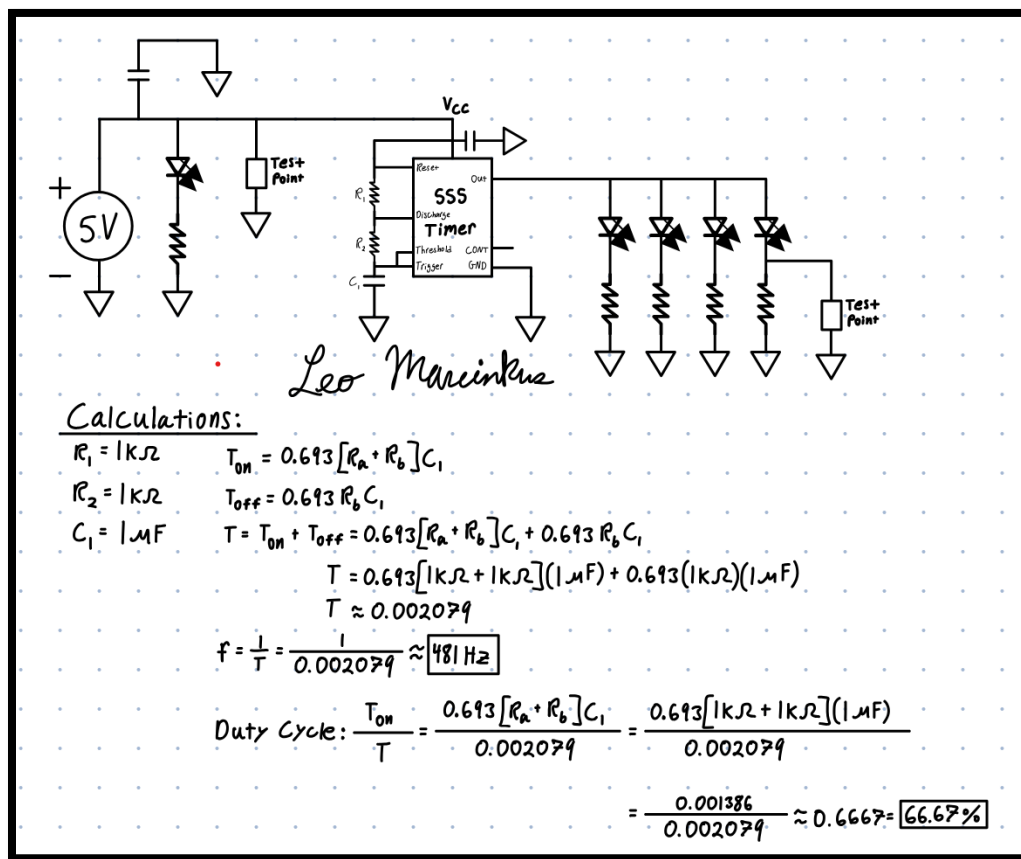


Figure 1: Napkin Sketch and Calculations

Schematic and PCB:

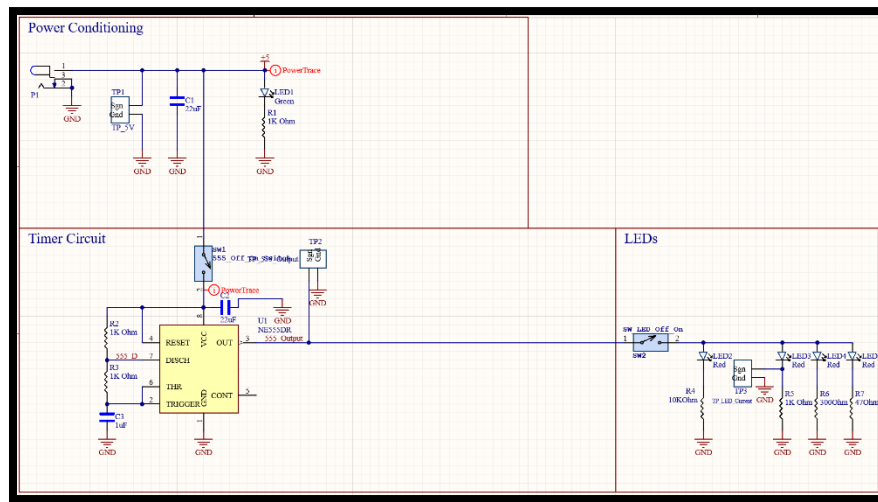


Figure 2: Altium Schematic

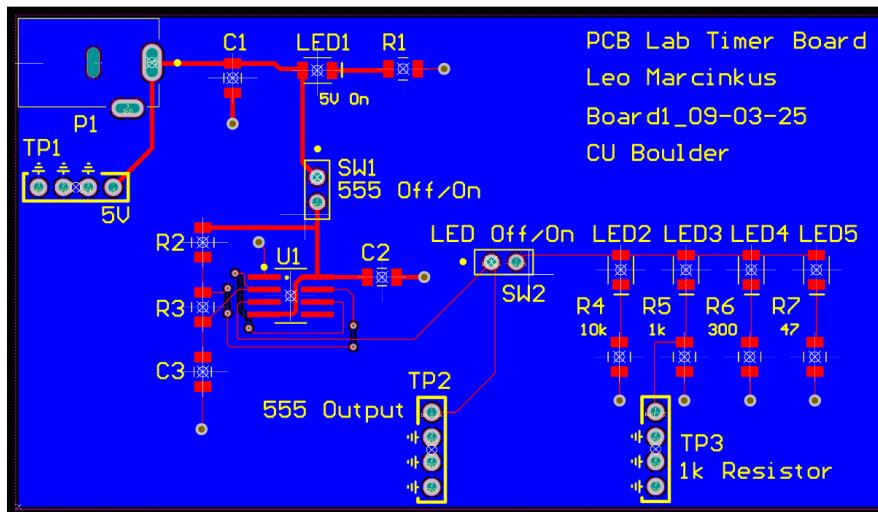


Figure 3: Altium PCB

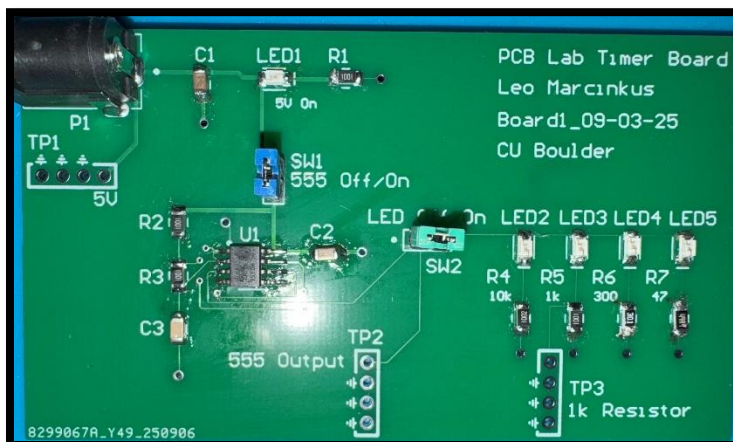


Figure 4: Completed Board 1

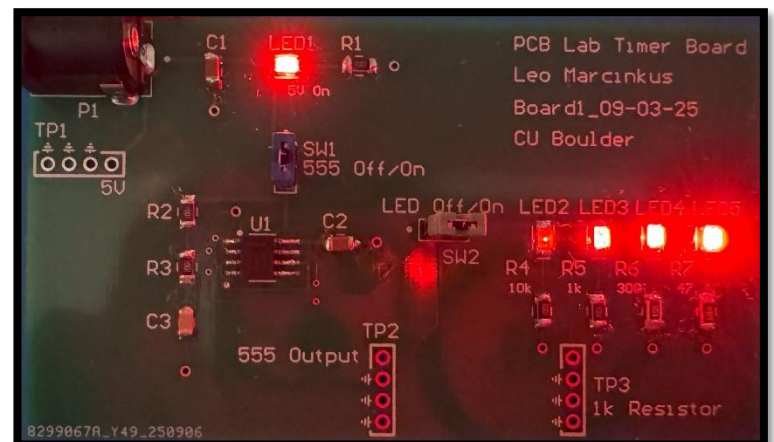


Figure 5: Powered Board 1

Oscilloscope Measurements:

Peak-to-peak voltage, frequency, duty cycle, and the rise/fall times were measured.

555 Timer output without load LEDs:

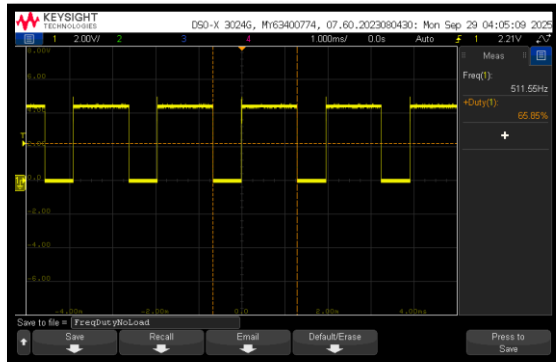


Figure 6: Frequency and Duty Cycle No Load

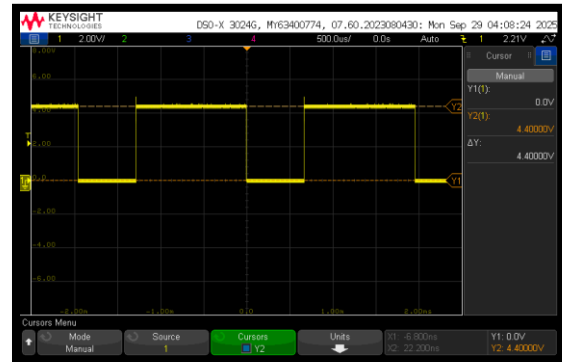


Figure 7: Peak-to-peak No Load

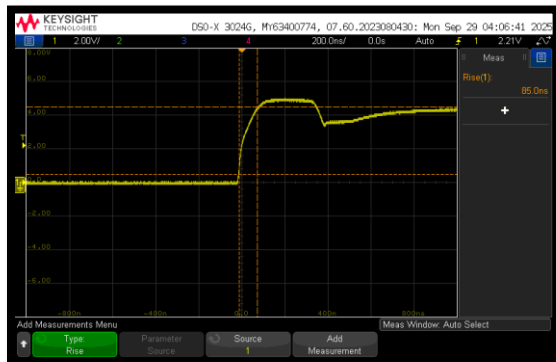


Figure 8: Rise Time No Load

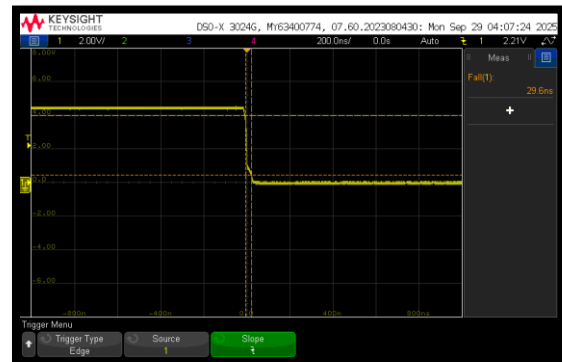


Figure 9: Fall Time No Load

The circuit's design should output $\sim 481\text{Hz}$ with a 66.67% duty cycle. Without a load, the oscilloscope measures the timer's output frequency to be $\sim 511\text{Hz}$ with a 65.85% duty cycle. The peak-to-peak voltage is 4.4V with a rise time of 85.0 ns and a fall time of 29.6 ns. These values perform close to what is expected.

555 Timer output with load LEDs:

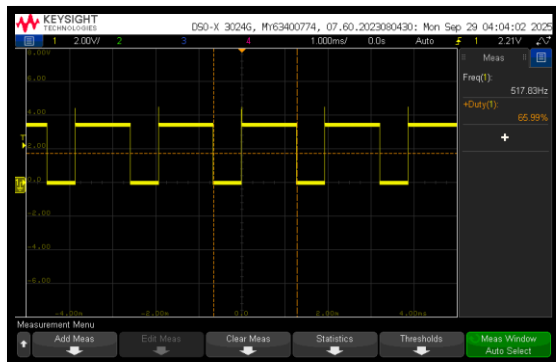


Figure 10: Frequency and Duty Cycle with Load

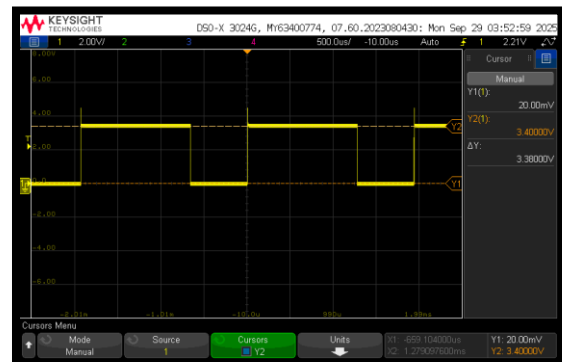


Figure 11: Peak-to-peak with Load

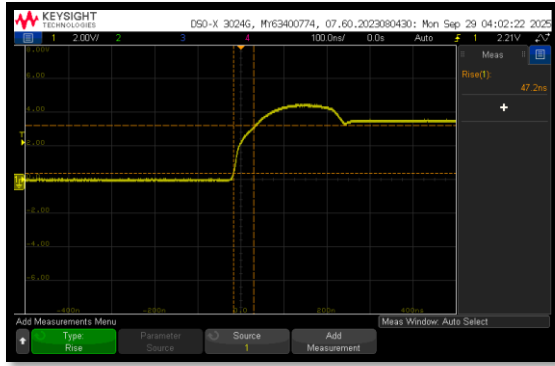


Figure 12: Rise Time with Load

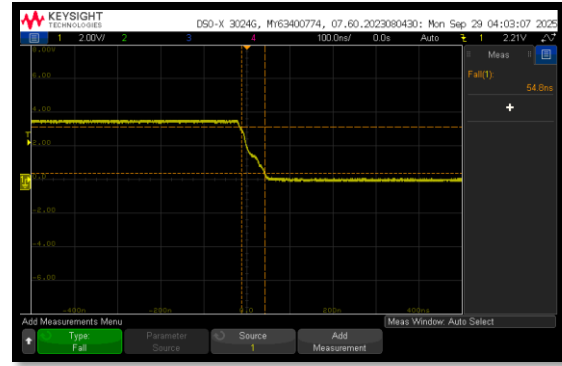


Figure 13: Fall Time with Load

With a load, the oscilloscope measures the timer's output frequency to be ~517Hz with a 65.99% duty cycle. The peak-to-peak voltage drops 1V to 3.4V and has a rise time of 47.2 ns with a fall time of 54.8 ns. These values perform close to what is expected with only the rise time being slightly quicker than expected. The next measurements are performed across load R5.

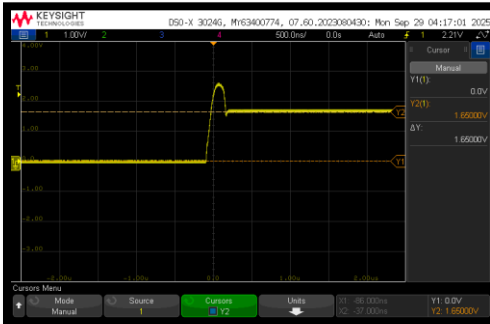


Figure 14: Peak-to-peak across R5

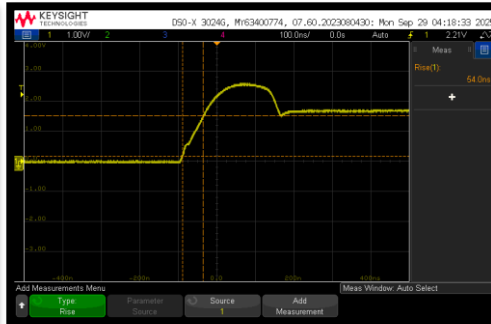


Figure 15: Rise Time across R5

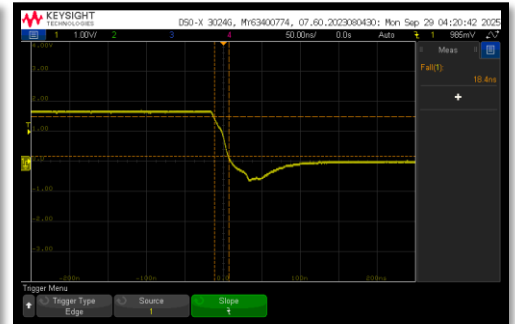


Figure 16: Fall Time across R5

The peak-to-peak across R5 is 1.65V, the rise time is 54.0 ns, and the fall time is 18.4 ns.

Thevenin Resistance LED Current Calculations:

$$V_{Open} = 4.4V, V_{Load} = 3, V_{LED} = 1.65V, I = V/R, I_{LED} = \frac{1.65V}{R}$$

The voltage measured across each of the LEDs is about ~1.65V. So, it will be used for each current calculation:

$$I_{10k\Omega} = \frac{1.65V}{10,000} = 0.165mA, I_{1k\Omega} = \frac{1.65V}{1000} = 1.65mA, I_{300\Omega} = \frac{1.65V}{300} = 5.5mA, I_{47\Omega} = \frac{1.65V}{47} = 35.106mA$$

$$I_{Load} \approx 42.4mA, R_{Th} = \frac{4.4V - 3.4V}{42.4mA} \approx 23.585\Omega$$

The Thevenin resistance is ~23.585Ω. Still a reasonable value for this circuit but a bit on the higher side.

Minimum Current for a Power Indicator LED:

The dimmest LED comes from the 10k Ω resistor with a ~66.6% positive duty cycle. There's about 0.165mA of current running through it. If the LED could glow at least 20% brighter, it would be easy to see and act as a good indicator LED. With a 100% positive duty cycle this would be ~0.236mA and should increase brightness about ~33%.

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Switching Noise:

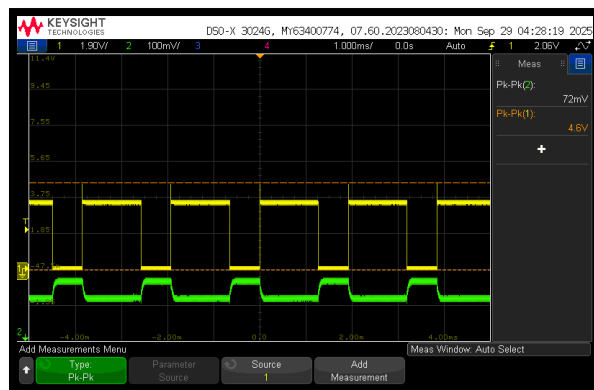


Figure 17: Power and Output

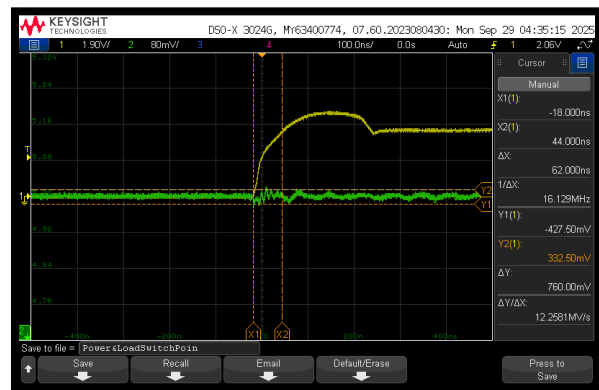


Figure 18: Switching noise and delay

The power rail noise is ~760mV, quite high, with a delay of 62 ns, also a bit high.

Good Design and Measurement Practices for Consistent Results:

1. Probe points are valuable for achieving accurate measurements. On such a small scale, placement on the circuit is not as important, but should still be considered as good practice.
2. Clear and consistent label names are crucial for being able to understand the circuit without having to constantly check back at the schematic.
3. Switches are useful to isolate power and data signals. This helps when recording data as measurements can be taken with and without a load, allowing for comparison.
4. Utilizing an oscilloscope is essential to achieve legible measurements.
5. Separating different sections of the circuit makes it easy to keep things organized. It is not necessary but helps when routing in Altium and is good practice.

Analysis:

1. *What worked well and should be carried over to the next board?*
 - a. Clear labels made assembly of the board easy as it was simple to tell what component should go where. Separating sections of the board significantly organized the design of the circuit. Including a power indication LED helped visualize what was happening after assembly. These practices, alongside inclusion of decoupling capacitors, will be carried into the next board design.
2. *What should be done differently?*
 - a. Although there were no issues, I think better component placement during the design phase would be beneficial. Some components during assembly were difficult to solder if another component was too close. I think including probe points on each LED would be beneficial to observe more load behavior as well.
3. *Any hard errors?*
 - a. No hard errors occurred preventing function.
4. *Any soft errors that can be corrected for the next board?*
 - a. No specific errors occurred when recording data. But including every component's value on the overlay of the board would be useful for the soldering process.

Conclusion:

Board 1 was successfully designed and assembled. The board utilizes a 5V powered 555-timer to produce a ~500Hz and ~66% duty cycle output. Four LEDs are powered, and each has its own resistor value to provide a different brightness. Measurements were taken to understand the behavior of this circuit with and without a load attached. Overall, I this board taught me good circuit design practices while also improving my soldering skills. Through the analysis of this board, the next design will feature multiple minor improvements.